

Research paper

An exploratory study on intelligent active cell balancing of electric vehicle battery management and performance using machine learning algorithms

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ABSTRACT

Battery Management Systems (BMS) rely on cell balancing to extend the longevity and efficiency of battery packs. Among these, active cell balancing techniques offer significant advantages over passive methods by redistributing charge more effectively, resulting in improved voltage balance, increased battery capacity, and an extended lifecycle. However, active cell balancing is often more complex, costly, and may introduce heat management challenges. Recently, machine learning (ML)-based active cell balancing controllers have emerged as a promising solution to optimize these systems. Leveraging dynamic charge control with probability distributions and advanced instrumentation, ML algorithms such as Predictive Analytic Recurrent Neural Networks (PA-RNN), Deep-Q Networks (DQN), Amortized-Q Networks (AQN), Adaptive Neural Networks (ADNN), and Automotive Controllers (AC) have demonstrated precise control over battery voltage, current, and State of Charge (SoC). These approaches significantly reduce SoC discrepancies, enhance energy efficiency, and mitigate common battery issues. Comparative analyses reveal that ML-driven active cell balancing not only outperforms traditional methods but also offers substantial improvements in performance, safety, and battery lifespan. This exploratory study highlights the disruptive potential of machine learning algorithms in revolutionizing Battery Management Systems, advancing both operational efficiency and overall battery health.

Acronyms

EVs	electric vehicles
BMS	Battery Management System
BESS	Battery Energy Storage System
SoC	State of Charge
PA-RNN	Predictive Analytic Recurrent Neural Networks
DQN	Deep-Q Networks
AQN	Amortized-Q Networks
ADNN	Adaptive Neural Networks
AC	Automotive Controllers
MPC	model predictive control

1. Introduction

Electrical vehicles (EVs) are very useful in tackling and reversing climate change and reducing greenhouse gas emissions. Regarding the environmental impact, EVs have a positive perspective, as they offer an opportunity that differs from gasoline-powered automobiles as energy sources for them are wind, solar and other renewable sources [1–4]. Another advantage that EVs hold is that they are environmentally friendly due to the reduced air pollution, translating to better health standards in society. Also, they assist in lowering demands for fossil energies, contributing to the bolstering of energy security. There are new advances in the battery technology of EVs and they are gradually moving into the category of lower costs and higher efficiency, which is also enhancing the rate of acceptance of EVs. Governments all over the world and the automotive industry are investing their money into

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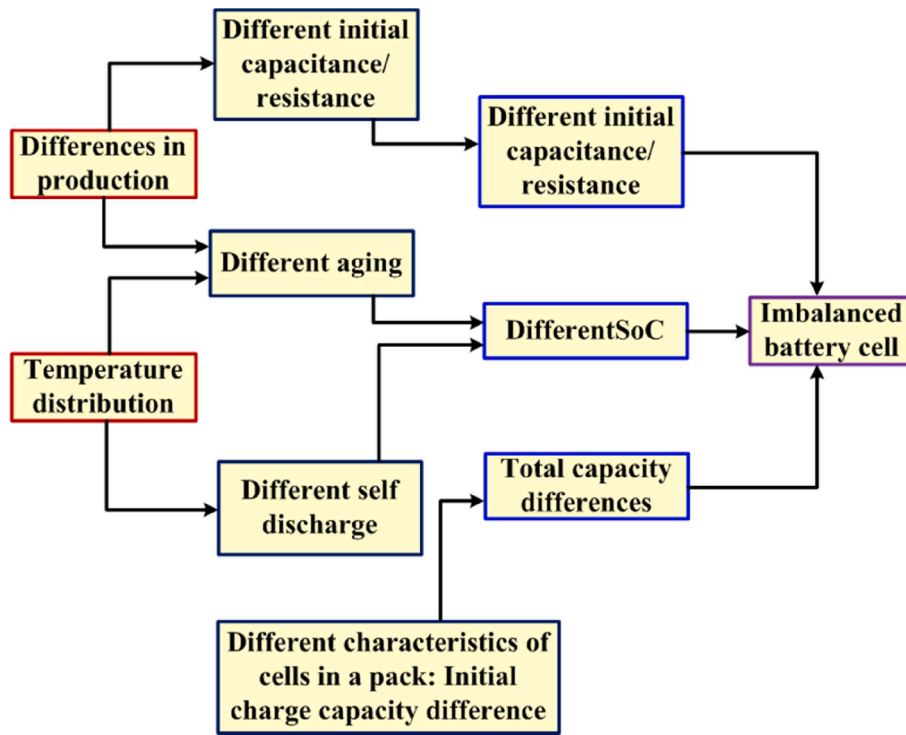


Fig. 1. Causes of cell imbalance.

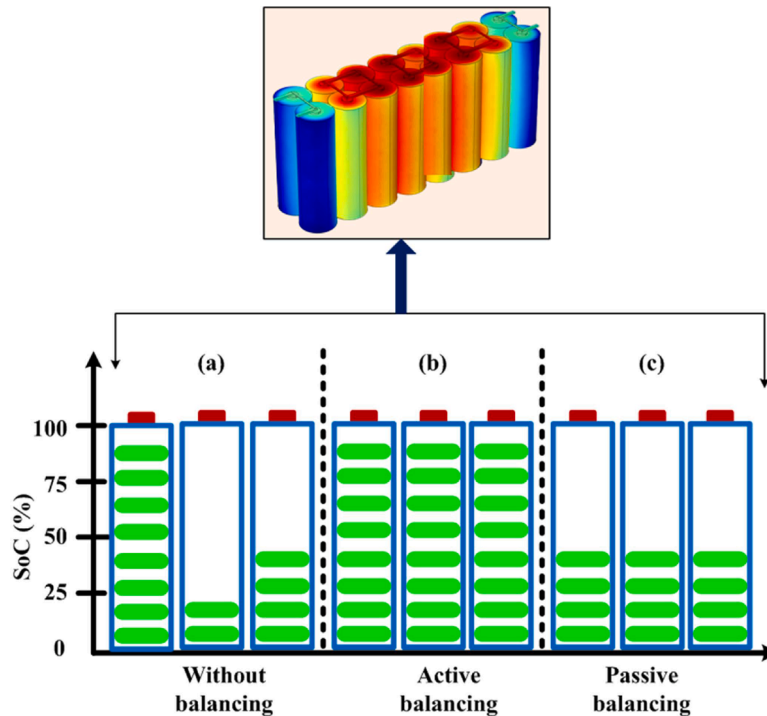


Fig. 2. Battery cell (a) Without balancing, (b) Active balancing, and (c) Passive balancing.

expanding the infrastructure of EVs since it plays a central position when it comes to the sustainable development of a green transportation system in future [5–8,43,46].

Energy storage systems play a very important role to balance the supply and demand as well as the reliability of conventional power sources [9]. Battery Energy Storage System (BESS) are still a developing industry and although Li-ion batteries are widely used, their adoption is not wide spread [45]. The battery capacity in electric vehicles provides

the best energy density and has a longer life span. It is noted that they are pivotal for EV performance as they define the range and efficiency of the equipment. Charging technologies target advancements to efficiency, costs, and charging time with an aim to make EVs more feasible. Battery Management System (BMS) in electric vehicle monitors and controls battery health, State of Charge (SoC), cell balancing and temperature to avoid condition such as over charge, deep discharge and thermal runaway. Regarding the choice of cell balancing in electric

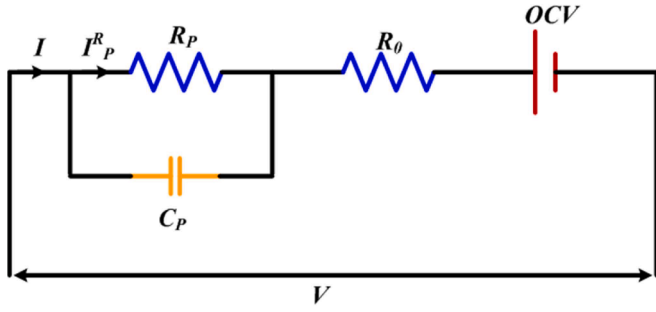


Fig. 3. Thevenin equivalent circuit.

vehicles, it means that it is selected to cause all the cells of the battery to charge and discharge in the same way.

This enables charging and discharging of batteries to be controlled in a way that would prevent the battery from overcharging or deep discharging which improves on battery life, capacity and reliability. Active balancing in which charge is dispersed over the cells and passive balancing where charges are dissipated via heat formation are techniques of cell balance in electric vehicle.

1.1. Research motivation and contribution

Recent advancements in machine learning (ML)-driven active balancing have shown promise in addressing these limitations. ML algorithms can dynamically optimize charge redistribution, reducing SoC imbalances, enhancing energy efficiency, and extending battery life. However, research in this area remains largely unexplored, with existing studies lacking comprehensive comparative analyses of ML-based techniques in real-world EV applications.

In this study, we propose an intelligent active cell balancing framework utilizing machine learning models, including PA-RNN, DQN, AQN, ADNN, and AC. The proposed system optimizes charge transfer in real-time, mitigating SoC imbalances while maintaining system stability. Through extensive simulations, we evaluate the effectiveness of these ML models and compare them with conventional balancing techniques.

Our findings demonstrate significant improvements in balancing efficiency, response time, and thermal stability, paving the way for enhanced BMS performance in next-generation EVs.

1.2. Highlights

- Presented an accurate solution for optimizing BMS through machine learning-based active cell balancing for electric vehicles.
- The PA-RNN (3.8A) minimizes SoC errors, Deep-Q (3.95A) dynamically adjusts control strategies, Amortized-Q (3.97A) supports quick decision-making, Adaptive NN (4.15A) provides high accuracy, and Automotive Controllers (4.32A) are tailored specifically for electric vehicle applications.

1.3. Organization of the manuscript

Section 2 presents the related work, Section 3 presents the modelling of Li-ion battery pack. Section 4 discusses various active balancing control systems. Section 5 covers control strategies in active cell balancing, and Section 6 concludes the study.

2. Related work

Battery management and cell balancing techniques are critical to ensuring the performance, longevity, and reliability of lithium-ion (Li-ion) batteries in electric vehicles (EVs). Several studies have explored different approaches to cell balancing, broadly classified into passive and active methods.

2.1. Traditional cell balancing techniques

A comprehensive review of cell balancing techniques is provided in [10], which discusses various battery models, SoC estimation methods, and performance management strategies. In addition, [11] evaluates the effectiveness of balancing circuits through convex optimization across multiple EV scenarios, demonstrating improvements in thermal balancing and internal resistance equalization. Further, [12] analyzes the causes of imbalances such as manufacturing variations, internal

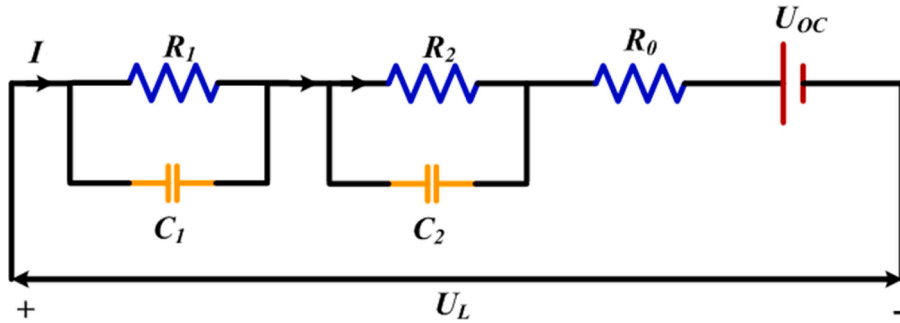


Fig. 4. Second order Thevenin equivalent circuit.

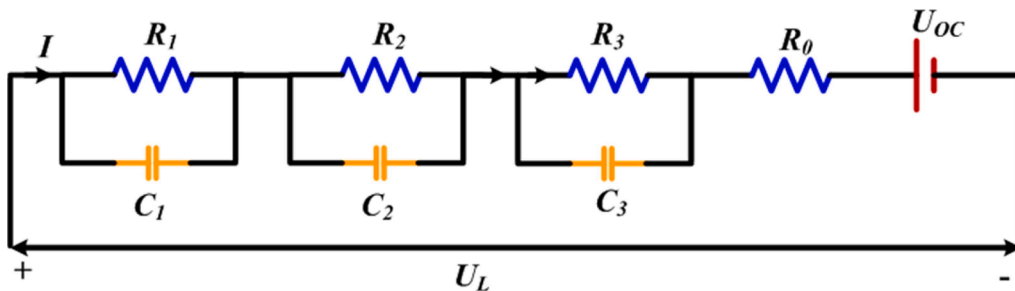


Fig. 5. Third-order Thevenin equivalent circuit.

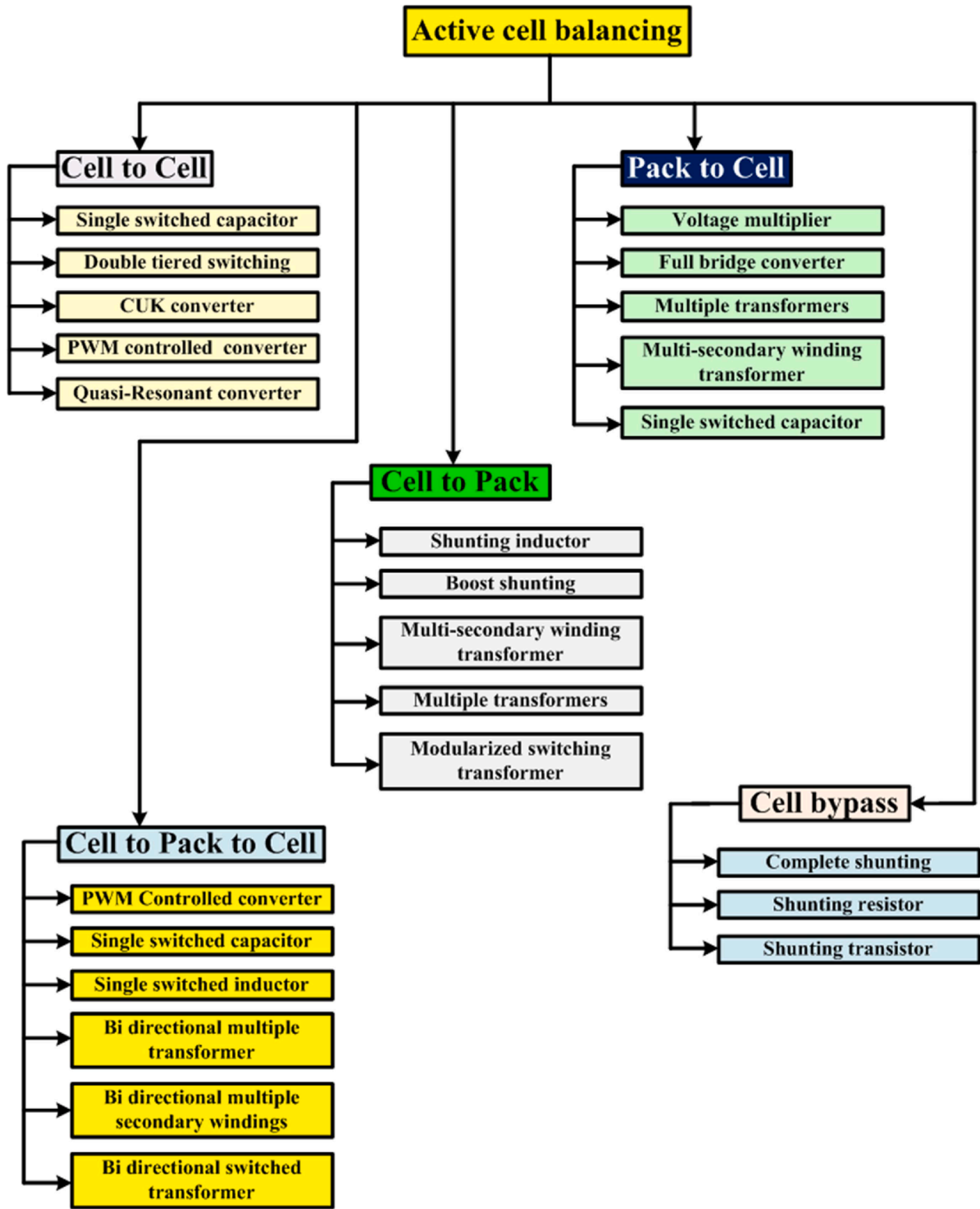


Fig. 6. Classification of active cell balancing.

resistance disparities, and inconsistent heating conditions—highlighting their impact on battery lifespan, power density, and temperature tolerance. Fig. 1 demonstrates an overview of causes of cell imbalance.

Several studies compare passive and active balancing techniques. Research in [13,14] confirms that active balancing is more effective than passive balancing for Li-ion batteries, leading to better energy efficiency

and longer battery life. Fig. 2 shows the balancing and misbalancing of battery cells. Studies such as [15–17] further examine different active balancing architectures, including DC-DC converter-based approaches, MATLAB-based performance comparisons, and power dissipation techniques to minimize SoC imbalances.

Table 1
Comparison between active cell balancing categories.

Category	Advantages	Disadvantages	Cost	Balancing Speed	Complexity in Control	Scalability	Reliability
Cell bypass [32]	- Efficient energy transfer - simple implementation - reduced power loss compared to passive balancing - high reliability	- Increased complexity with more cells - higher initial cost due to additional components - potential heat generation	Moderate to high	Moderate to fast	Requires control circuitry to manage to switch actions	Good scalability for medium to large battery packs	High, as it uses fewer components than other active methods.
Cell to Cell [33]	- High efficiency - Balances cells regardless of SOC - Improves battery performance - Lifespan - Better capacity utilization	- High complexity - Increased cost - Larger and heavier system	Higher due to sophisticated components and circuitry, varies with complexity and cell count.	Faster than passive methods, can address imbalances quickly during charge/discharge cycles.	Requires sophisticated control algorithms, robust monitoring and feedback systems.	Can be scaled for large battery packs but increases in cost and complexity.	Typically more efficient than passive balancing, leading to better performance and energy savings.
Pack to Cell [34]	- Improved Efficiency - Enhanced Lifespan - Better Performance - Increased Safety	- Higher Cost - Increased Complexity - Energy Consumption	Generally higher due to the need for active balancing circuitry and control systems.	Typically faster than passive balancing, ensuring quicker equalization of cell voltages.	Requires advanced algorithms and precise control mechanisms to manage balancing effectively.	Suitable for small to medium-sized battery packs. Scaling to larger packs may increase complexity and cost.	High reliability when properly implemented, but reliability can be compromised by component failures.
Cell to Pack [35]	- Efficiency - Improved Performance - Longer Lifespan - Enhanced Safety	- Complexity - Cost - Efficiency Loss	Initial setup costs are high due to the need for specialized hardware and control systems.	Typically faster than passive balancing, as active systems actively transfer charge between cells.	Control algorithms must be sophisticated to manage balancing operations effectively without compromising performance.	Can be scaled to accommodate various pack sizes and chemistries by adjusting the number and type of balancing circuits.	Active balancing enhances pack reliability by ensuring uniform cell performance and reducing the risk of individual cell failure.
Cell to Pack to Cell [36]	- Simplified BMS design - Higher energy density - Improved cell balancing accuracy - Potential for extending battery pack lifespan	- Complex cell grouping - Potential for uneven pack ageing - Increased complexity in BMS design - Higher cost due to additional components - Space and weight constraints in the pack	Low	Moderate to Fast	Moderate to High	Moderate	High

2.2. Innovative active balancing strategies

Several advanced hardware-driven balancing approaches have been proposed. A rapid cell-to-cell balancing circuit using a push-pull converter, relays, and a battery-monitoring IC is discussed in [18], achieving 89.5 % efficiency at low implementation costs. The direct energy transfer method in [20] improves balancing by reducing sensor usage through simultaneous observation techniques, verified through both simulations and experimental results. Meanwhile, the cloud-based smart BMS framework in [19] integrates cloud computing to enhance real-time decision-making and improve computational efficiency.

Refs. [21,22] analyze the impact of active balancing on battery lifespan and efficiency, demonstrating a 10 % improvement in system life when dynamic controllers are incorporated. Meanwhile, [23] focuses on reducing the computational burden in active balancing, enabling large-scale implementation by optimizing transfer phase dynamics and error control techniques. Studies like [24,25] present fuzzy-based balancing circuits and buck-boost equalizers, showing improvements in SoC stabilization, energy efficiency, and reduced switching requirements.

2.3. Machine learning-based balancing approaches

Recent research has explored the integration of artificial intelligence (AI) and machine learning (ML) techniques to enhance battery balancing. The work in [27] highlights the role of AI-driven predictive health management for diagnosing battery degradation. Similarly, [28] presents an AI-based charge redistribution approach, optimizing battery capacity while minimizing inconsistencies in SoC levels.

Despite these advancements, existing studies lack a comprehensive comparison of machine learning-based balancing techniques and their real-world impact on battery management. To bridge this gap, this study proposes a ML-driven active cell balancing framework, leveraging the strengths of:

- PA-RNN – Optimizing SoC balancing with minimal errors.
- DQN – Dynamically adjusting balancing control strategies.
- AQN– Supporting real-time decision-making.
- ADNN – Providing higher precision in charge redistribution.
- AC – Tailored for EV applications to enhance BMS efficiency.

This study systematically evaluates these models against conventional methods, demonstrating improvements in balancing accuracy, response time, and battery lifespan.

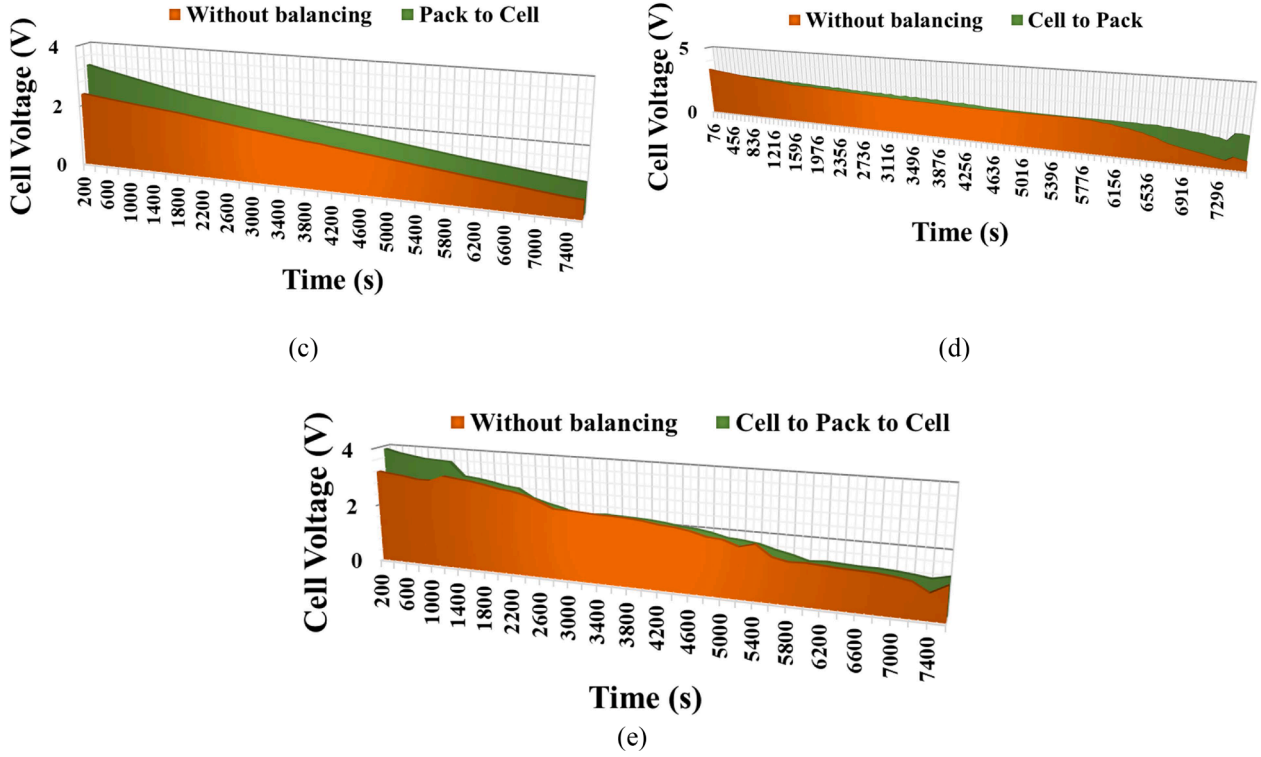


Fig. 7. Operational characteristics of active cell balancing for (a) Cell bypass, (b) Cell to Cell, (c) Pack to Cell, (d) Cell to Pack, and (e) Cell to Pack to Cell.

3. Equivalent circuit model

3.1. One RC-modelled Li-ion battery

By simulating the voltage response to current flow in a battery using a single resistor and capacitor, the 1RC model simplifies the dynamics of the battery. It provides computational efficiency without delving into internal chemical processes, unlike electrochemical models. However, it can lack accuracy in capturing intricate electrochemical behaviours. A representation of the Thevenin-modelled 1RC simulated Li-ion battery can be seen in Fig. 3. The Thevenin equivalent circuit model comprises essential components such as ohmic resistance R_0 , an RC network with polarization resistance R_p and polarization capacitance C_p , and the open-circuit voltage (V_{OC}). These components interact to describe battery behaviour. The following is a description of the relationships that exist between these components [29,43,44]:

$$\frac{dI^{Rp}(t)}{dt} = \frac{I(t) - I^{Rp}(t)}{R_p C_p} \quad (1)$$

$$U(t) = R_p I^{Rp}(t) + R_0 I(t) + V_{OC}(t) \quad (2)$$

The polarization resistance current, represented as I^{Rp} , is positive while charging and negative when discharging, and the terminal voltage, U , is represented by the symbol. The nonlinear relationship between SoC and open-circuit voltage (V_{OC}) is used to determine the voltage in the following way:

$$V_{OC}(t) = f(\text{SoC}(t)) \quad (3)$$

$$\text{SoC}(t) = \text{SoC}(t_0) - \eta_c \int_{t_0}^t \frac{I(\tau)}{Q_c} d\tau \quad (4)$$

This function f depicts the unknown nonlinear connection between open-circuit voltage (V_{OC}) and SoC. $\text{SoC}(t)$ and $\text{SoC}(t_0)$ represent the SoC values at time t and beginning time t_0 , respectively. The Li-ion battery's maximum capacity is Q_c and the coulomb efficiency (η_c) is around 1.

3.2. Two RC-modelled Li-ion battery

The Second-order Thevenin equivalent circuit model reflects battery polarization dynamics [30]. In Fig. 4, the model includes V_{OC} as the open-circuit voltage, R_0 as internal resistance, I as charge/discharge current from the voltage source, U_1 and U_2 as first and second-order RC network voltages, and U_L as terminal voltage. R_1 and R_2 represent electrochemical and concentration polarization resistance, respectively. C_1 and C_2 represent electrochemical and concentration polarization capacitance, respectively. The battery is mathematically represented in Eq. (5).

$$\left. \begin{aligned} U_L &= U_{OC}(\text{SoC}) - i(t)R_0 - U_1 - U_2 \\ \frac{dU_1}{dt} &= -\frac{U_1}{R_1 C_1} + \frac{i}{C_1} \\ \frac{dU_2}{dt} &= -\frac{U_2}{R_2 C_2} + \frac{i}{C_2} \end{aligned} \right\} \quad (5)$$

Select $[\text{SoC}, U_1, U_2]$ as state variables. Eq. (6) expresses battery SoC after discretization.

$$\left[\begin{array}{c} \text{SoC}_{k+1} \\ U_{1,k+1} \\ U_{2,k+1} \end{array} \right] = \left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 - T/\tau_1 & 0 \\ 0 & 0 & 1 - T/\tau_2 \end{array} \right] \left[\begin{array}{c} \text{SoC}_k \\ U_{1,k} \\ U_{2,k} \end{array} \right] + \left[\begin{array}{c} \eta_c Q_c T \\ T \\ T \end{array} \right] i_k \quad (6)$$

$$U_{L,k+1} = U_{oc}(\text{SoC}, k+1) - U_1 - U_2 - IR_0$$

3.3. Three RC-modelled Li-ion battery

The equations that describe the dynamics of the voltage can be stated as follows in a third-order RC network model for a lithium-ion battery model [30]:

Table 2
Control variables in active cell balancing.

Control variable	Advantages	Disadvantages	Usage application
Voltage Difference	- Simple to implement - Effective for balancing SoC	- May not address capacity differences - Can cause over-balancing in cells with different capacities	- EVs - Consumer electronics - Renewable energy storage
SoC	- Balances cells based on the actual charge - Improves overall battery pack performance	- Requires accurate SoC estimation - Complex SoC algorithms	- EVs - Stationary energy storage systems - Aerospace applications
Temperature	- Ensures thermal management and safety - Prevents thermal runaway - Extends battery life	- Additional sensors and complexity - Balancing may be slower due to thermal constraints	- High-performance applications - Energy storage systems - EVs
Battery current	- Dynamic balancing during charging/discharging - Can improve charge/discharge efficiency	- Requires sophisticated control algorithms - Increased system complexity	- EVs - Renewable energy storage - Industrial battery systems
Battery impedance	- Addresses cell ageing and health - Can predict and prevent cell failures	- Requires complex impedance measurement techniques - Higher cost due to advanced measurement equipment	- Long-life applications (e. g., grid storage) - Medical devices - Aerospace applications
Battery power	- Optimizes power distribution across cells - Enhances performance during high power demand	- High computational requirements - Complexity in real-time power management	- EVs - High-performance applications - Industrial applications

$$\left. \begin{aligned}
 U(t) &= V_{OC}(SoC(t)) - I \cdot R_0 - U_1(t) - U_2(t) - U_3(t) \\
 \tau_1 \frac{dU_1(t)}{dt} &= \frac{U(t) - U_1(t)}{R_1} \\
 \tau_2 \frac{dU_2(t)}{dt} &= \frac{U_1(t) - U_2(t)}{R_2} \\
 \tau_3 \frac{dU_3(t)}{dt} &= \frac{U_2(t) - U_3(t)}{R_3}
 \end{aligned} \right\} \quad (7)$$

There is a correlation between the amount of complexity and precision necessary to capture the dynamic behaviour of a Li-ion battery and the applicability of RC, 2RC, and 3RC network models for modelling Li-ion batteries. Only a single resistor (R) and capacitor (C) are required for the RC Network, which is the most basic type. However, it is not accurate enough to capture complicated electrochemical dynamics, therefore it is only suitable for basic analysis and rapid simulations. A better description of polarization effects and transient behaviour is made possible by the inclusion of a second RC branch to the 2RC Network. It is appropriate for more extensive simulations and analysis since it offers better accuracy in comparison to the RC model. Higher-order polarization effects are included in the 3RC Network as shown in Fig. 5, which incorporates a third RC branch to provide even more realism in the process of collecting battery dynamics. Among the three models, it provides the best degree of accuracy; nevertheless, it comes with more complexity and requires more computing.

Table 3
Computational demands and optimization strategies for BMS.

Algorithm	Computational Demands	Optimization Strategies for BMS
PA-RNN	High demand due to complex recurrent structures and sequential data processing. Requires significant memory and processing power to handle time-series data.	1. Pruning: Reduce the number of neurons in the network. 2. Model Quantization: Use lower precision for weights. 3. Network Distillation: Use a smaller network to approximate a larger one.
DQN	High computational complexity due to deep neural networks and Q-learning computations. Involves large state-action spaces.	1. Experience Replay Optimization: Use prioritized experience replay to speed up learning. 2. Simplifying Network Architecture: Reduce the number of layers and neurons. 3. Target Network Frequency Adjustment: Update target networks less frequently to save computation.
Amortized-Q Networks	Computationally intensive due to the need for multiple Q-function estimations. Involves deep reinforcement learning tasks.	1. Action Pruning: Limit the number of possible actions to reduce state-action space. 2. Parallelization: Distribute training across multiple smaller devices. 3. Offloading Computations: Offload some calculations to a more powerful processor if feasible.
ADNN	Moderate to high computational demand due to the need for dynamic adjustments to the network during training.	1. Adaptive Learning Rates: Reduce complexity by adjusting learning rates during training. 2. Sparse Representations: Use sparsity in weight matrices to reduce computation. 3. Incremental Training: Train the model incrementally to avoid the overhead of large-scale batch processing.
AC	Computational demands depend on the complexity of the control algorithms, which may require real-time decision-making with low latency.	1. Model Simplification: Use simpler models (e.g., linear regression) for control tasks. 2. Efficient Optimization Algorithms: Use methods like model predictive control (MPC) to reduce computation. 3. Hardware Acceleration: Use specialized hardware accelerators like FPGA or ASICs to speed up real-time control.

4. Active cell balancing

In large battery systems, such as those used in electric vehicles and energy storage systems, active cell balancing is a crucial strategy for regulating SoC throughout the cells in a battery pack [31]. This is especially true for big battery packs. To maximize the battery pack's efficiency, longevity, and safety, the objective is to make certain that all of the cells keep their charge levels equal. As shown in Fig. 6, the active cell balancing strategies can be subdivided into (i) Cell to Cell, (ii) Pack to Cell, (iii) Cell to Pack, (iv) Cell bypass, and (v) Cell to Pack to Cell. The sections of these five strategies are involves the power electronic converters, active elements, passive elements and transformers (Table 1).

Different approaches are utilized to redistribute charge among battery cells, each operating at specific voltage levels to enhance balancing performance:

Table 4

Machine learning-based equalizer controllers.

Equalization controller	Converter	Variables used	Contribution	Limitation	Objectives
PA-RNN [39]	- Buck converters - Dual active bridge (DAB)	- Current stress - Switching frequency - Input voltage - Output voltage - Load current - Switching losses	- Innovative PA-RNN model - Data-light and on-call predictions - Enhanced performance evaluation	- Complexity in customization - Scalability to different converter types - Retraining for different operating conditions	- Increase AI penetration in power electronics - Optimize modulation techniques - Enhance accuracy and efficiency - Provide practical solutions - Validate through experiments
Deep Q-Network [40]	- Grid-forming Converters - Grid-following Converters	- Energy-mix proportion - System load - Short circuit level - Load fluctuations	- Deep Reinforcement Learning (DRL) Framework - Stability and sustainability - Data-driven estimation	- Intermittency of renewable energy sources - Complexity of real-time implementation - Data requirements	- Optimize energy generation - Ensure power system stability - Improve system sustainability - Validate DRL applicability
Amortized Q-Learning [41]	Extended Phase Shift DAB	- SoC - Voltage (V) - Current (A) - Temperature (K)	- High efficiency - Isolation - High power density	- Complexity - Cost	- Efficient energy transfer between cells to equalize SoC. - Minimize energy losses and enhance the overall efficiency of the BMS. - Ensure safe operation through isolation and precise control.
Adaptive Neural Network [42]	Bidirectional buck-boost converter	- SoC - Voltage (V) - Current (A) - Temperature (K)	- Reduced ripple - Improved thermal management - Scalability	- Control complexity - Synchronization	- Reduce current ripple and enhance the thermal management of the battery pack. - Scale efficiently for different power levels without significant redesign. - Optimize the balance of cells dynamically using real-time data.
Automotive controller [43]	Joint Buck-Boost (BB) and Switched-Capacitor (SC) converters	- Balancing current - Voltage - Balancing speed - Efficiency - Cell count	- AC-AC equalization mode - High balancing speed and efficiency - Compact size and low cost - Analytical verification	- Complexity - Scalability - Implementation cost	- Efficient voltage equalization - Maintain compact design - Verify performance - Enhance balancing modes - Comparative analysis

- Cell Bypass Balancing (3.4 V): Maintains voltage uniformity by diverting excess energy from overcharged cells, preventing over-voltage conditions (Fig. 7a).
- Cell-to-Cell Balancing (3.2 V): Facilitates direct charge transfer between adjacent cells to equalize their voltages, enhancing overall pack stability (Fig. 7b).
- Pack-to-Cell Balancing (3.5 V): Transfers excess energy from the battery pack to individual cells with lower SoC, ensuring better distribution (Fig. 7c).
- Cell-to-Pack Balancing (3.6 V - 3.7 V): Allows overcharged cells to transfer surplus charge back to the pack, which can then be used to power other low-SoC cells (Fig. 7d).
- Cell-to-Pack-to-Cell Balancing (3.9 V): A hybrid method that combines pack-level and cell-level balancing, redistributing energy from high-voltage cells to the pack and then to lower-voltage cells for improved efficiency (Fig. 7e).

These intelligent balancing methods, integrated with machine learning algorithms, enhance the performance of BMS by dynamically adapting to variations in charge levels, thus optimizing battery health and reliability.

5. Control strategies in active cell balancing

Active cell balancing is a complex technology used in BMS to maintain the same SoC for all cells in a battery pack, improving performance and lifespan. This approach uses control mechanisms to transfer energy from higher to lower-charged cells. The main active cell balancing factors are SoC, voltage, current, temperature, and capacity [37]. SoC is crucial to active cell balance. It shows cell current charge compared to capacity. The BMS needs an accurate SoC estimate to know

when and how much to balance each cell. Algorithms that track voltage, current, and temperature over time estimate SoC. To avoid cell damage from overcharging or deep discharging, the SoC should be equalized across all cells. Another important control variable is voltage. The BMS detects cell voltage differences by monitoring each cell. Imbalances may be detected via voltage discrepancies. Cells with greater SOC transmit energy to those with lower voltages during balancing. This voltage-based balancing keeps all cells within their ideal voltage range, improving battery pack safety and efficiency [38].

Current regulation is crucial for cell energy transfer. Active balancing circuits transfer charge using inductors, capacitors, or transformers [26]. The current-controlled energy transfer rate must be optimized to balance cells without thermal or electrical stress. Controlling current during balancing prevents overheating and component wear. Voltage, temperature, SoC, current flow, balancing time, energy transfer rate, and balancing strategy types are the control variables listed in Table 2 for active cell balancing. Table 2 represents the key control variables in active cell balancing, highlighting their advantages, disadvantages, and usage applications. Voltage difference and SoC-based balancing are widely used due to their simplicity and effectiveness, but they require accurate estimation and may not account for capacity variations. Temperature, battery current, impedance, and power also play critical roles in ensuring safety, efficiency, and long-term battery health, especially in applications like EVs, renewable energy storage, and aerospace systems.

Battery management relies on temperature for balance and cell health. Cells behave differently at various temperatures, causing imbalance. BMS dynamically adjusts balancing technique based on temperature. If a cell overheats, energy transfer rates may be decreased to minimize thermal runaway and prolong battery life. Although capacity is not directly regulated, it affects the balancing method. Different

Table 5
Control strategy of machine learning-based active cell balancing.

Control strategy	Advantages	Challenges	Description
PA-RNN	- Adaptive to changing battery conditions. - Predictive capability enhances balancing accuracy. - Can handle nonlinear behaviours effectively.	- Computational complexity due to real-time data processing. - Initial setup and training require significant data and computation resources. - Requires robust validation in real-world battery systems.	- Uses particle dynamics to model and predict cell behaviour, contributing to dynamic balancing decisions. - Processes temporal data from PA to learn and optimize balancing strategies over time.
Deep Q-Network	- Adaptive to varying battery conditions. - Can optimize balancing strategy over time.	- Requires extensive training data or simulations. - Computationally intensive for real-time applications.	- Integration with real-time Battery Management Systems. - Ensuring safety and reliability during operation.
Amortized Q-Learning	- Learns optimal balancing actions based on rewards and penalties. - Reduces the need for predefined rules. - Can handle complex systems.	- Requires initial training phase with extensive data. - Training and updating Q-values can be resource-intensive. - Delays in response during initial learning phases.	Utilizes Q-learning, a form of reinforcement learning, to optimize cell balancing decisions over time.
Adaptive Neural Network	- Adaptive to changing battery conditions. - Can handle nonlinear relationships and complex dynamics. - Improves accuracy and efficiency over time with learning. - Reduces manual tuning and dependency on predefined rules.	- Requires training data and initial model development. - Computationally intensive, requiring real-time processing. - Complexity in model architecture and parameter selection. - Potential overfitting if not properly regularized.	Utilizes artificial neural networks to dynamically adjust balancing actions based on real-time battery data.
Automotive controller	Adjusts balancing parameters in real-time based on feedback from the battery system and operational conditions.	Responsive, adapts to battery ageing	Complex algorithms, continuous monitoring required

cell capacities decay differently, causing imbalances. Capacity information helps the BMS balance cells with lesser capacities and prevent overcharging or discharging. This helps extend battery pack life. These variables are managed using advanced algorithms and control mechanisms. Balancing is optimized through robust model predictive control, dynamic control, and fuzzy logic control. These algorithms employ

prediction models and real-time sensor data to achieve optimum balance. Table 3 compares computational demands and optimization strategies for various BMS algorithms. High-complexity methods like PA-RNN, DQN, and Amortized-Q Networks require techniques like pruning, parallelization, and hardware acceleration to improve efficiency. Moderate-demand algorithms like ADNN and AC benefit from adaptive learning rates and model simplifications, ensuring real-time decision-making in battery management.

Real-time data capture and processing make active cell balance variable control difficult. The BMS must continually monitor and change balancing operations depending on battery pack dynamics. High-precision sensors, strong communication networks, and powerful CPUs are needed for this hardware and software integration. Active cell balancing requires accurate SoC, battery voltage, battery current, battery temperature, and battery capacity management. Effective control of these factors ensures that all battery pack cells work together to maximize performance, safety, and longevity. The advantages of active cell balancing in battery health and efficiency justify its complexity and expense.

5.1. Software-based active cell balancing

Active cell balancing is a software-based technique that uses algorithms to regulate battery cell equalisation. This technique optimises charge and discharge rates, resulting in increased efficiency and lifetime. Table 4 Machine learning-based equalizer controllers: PA-RNN (Predictive Analytic Recurrent Neural Network), Deep-Q Network, Amortized-Q, Adaptive Neural Network (Adaptive NN), Automotive Controller. These controllers optimize battery cell balancing through advanced machine learning algorithms.

5.2. Control strategy in active cell balancing

The exact application requirements, such as the chemistry of the battery, the structure of the pack, and the operating environment, determine which of these control systems presents a unique set of benefits and problems. Several criteria, including performance objectives, the level of complexity that can be tolerated, and the resources that are available for installation and maintenance, are often taken into consideration when selecting a control approach. Battery systems across a variety of sectors, such as the automotive industry, renewable energy, and portable electronics, are seeing an increase in efficiency, dependability, and safety as a result of advancements in control algorithms and technologies. These advancements are driving advances in active cell balancing. Table 5 shows the control method for active cell balancing based on machine learning. This strategy uses predictive algorithms to increase efficiency, minimise faults in SoC, optimise cell voltage distribution, and adapt to different situations.

Cell voltage, SoC, temperature, current, energy transfer rates, and balancing time are important factors. Methods use up-to-the-minute information to fine-tune balancing operations for improved performance and battery pack life. Battery cell voltages are depicted in Fig. 8 (a) Without balancing, (b) PA-RNN, (c) Deep Q-Network, (d) Amortized Q-Learning, (e) Adaptive Neural Network, and (f) Automotive Controller.

PA-RNN technique uses recurrent neural networks to predict battery performance and behaviour over time. PA-RNN models are very useful for predicting battery charge-discharge cycles, which is critical for optimizing BMS. The PA-RNNs can maintain accuracy when scaled to bigger battery packs because of their capacity to learn temporal connections between battery data sequences. However, the computational demand increases as the model complexity increases with pack size. Real-world events, such as changing temperatures and usage patterns, generate noise that can impair the accuracy of PA-RNN unless it is regularized or fine-tuned to specific settings. PA-RNNs must be trained on a variety of data sets that cover these circumstances in order to

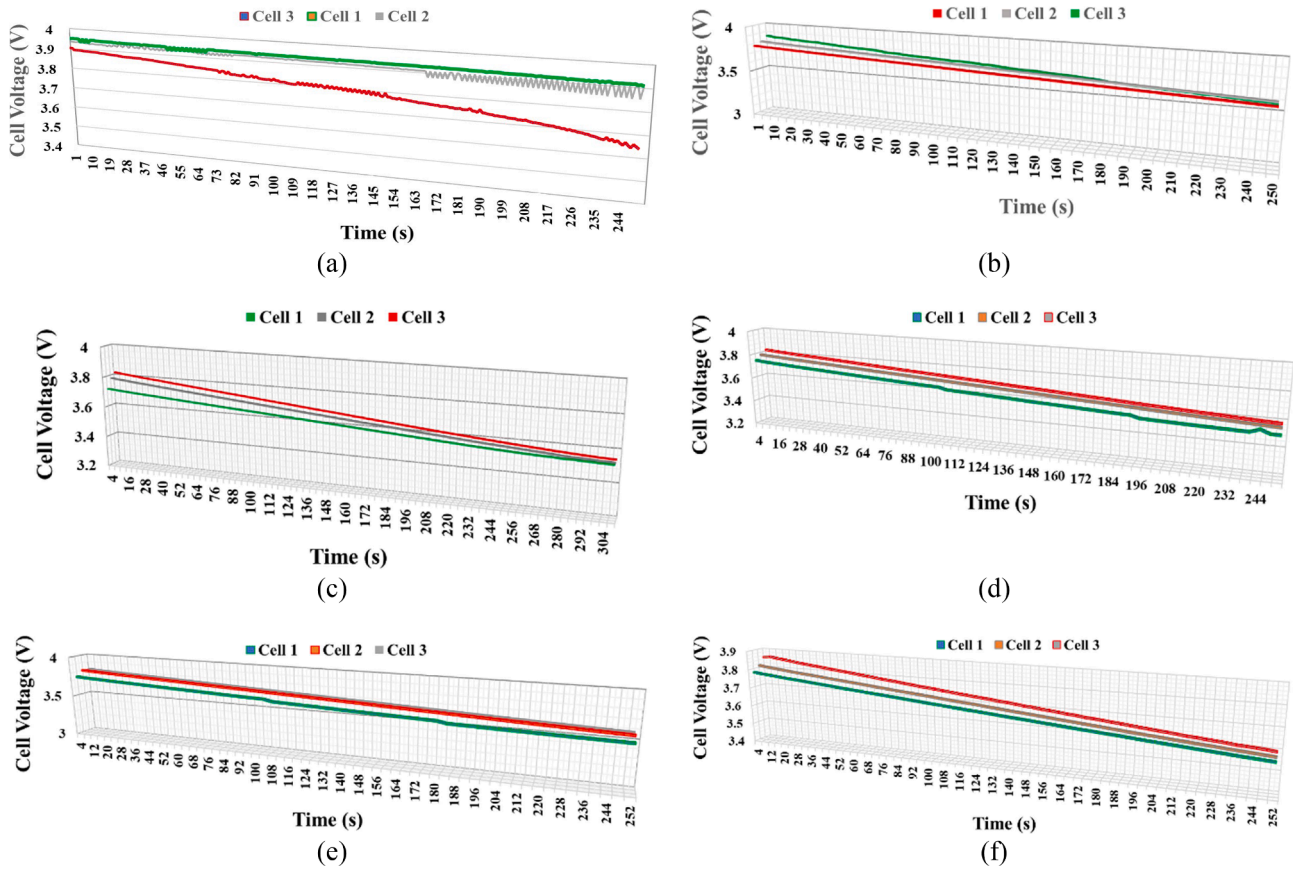


Fig. 8. Battery cell voltage for (a) Without balancing, (b) PA-RNN, (c) Deep Q-Network, (d) Amortized Q-Learning, (e) Adaptive Neural Network, and (f) Automotive Controller.

generalize successfully. DQN is a reinforcement learning method that is widely used to optimize decision-making in energy systems, such as massive battery pack charging and discharging cycles.

DQN main benefit is its capacity to learn optimum policies from interactions with the environment, which makes it flexible to real-world settings. Scaling DQN for bigger battery packs necessitates a larger volume of training data and more processing resources, which may cause slower convergence. DQN may suffer in circumstances with temperature fluctuations and diverse usage patterns unless the model is trained under these unique parameters. Furthermore, regular updates and fine-tuning are required to maintain the resilience of the policy learnt by DQN in these dynamic situations. A-QN solves the scalability problem of standard Q-learning approaches, such as DQN, by effectively approximating the Q-value function using neural networks. This is especially beneficial for large-scale battery systems, where precise Q-values may be computationally expensive. Amortized-QN outputs Q-values from a single neural network, which reduces memory and processing needs.

Its ability to scale to bigger battery packs outperforms existing approaches. A-QN, like DQN, requires considerable training on a variety of real-world events. Temperature and usage patterns remain challenging since the model must explicitly account for these parameters to avoid overfitting or underfitting. ADNNs are intended to adapt to changing surroundings and are especially useful in dynamic systems such as battery management, where operating circumstances can change quickly. They excel in fine-tuning battery models based on fresh data, resulting in improved prediction and control.

Scaling ADNN to bigger battery systems is possible, but the problem is adjusting to varied real-world situations such as temperature changes and varying charging/discharging rates. The ADNN may be extremely successful when regularly updated with real-world data, but caution

must be exercised to minimize model performance drift owing to unexplained factors. AC systems are often incorporated with vehicle BMS, with the goal of maximizing battery performance in real time. The growing number of control parameters makes scalability in automobile controllers for bigger battery packs a significant issue. In actual applications, Automotive Controllers must account for changing temperatures and use patterns, such as quick charging or high-load circumstances, which can have an influence on battery performance.

While AC systems are ideal for real-time operations, they necessitate sophisticated models and control algorithms that can handle a wide range of environmental variables without jeopardizing battery longevity or safety. Battery cell voltage is one of the measuring tools to know the accuracy of active cell balancing. Initially, under unbalanced conditions, the battery cell voltages are dislocated and each cell voltage is not maintained properly. Such as cell 1, cell 2 and cell 3 behaviour is clearly described in Table 6.

There is the potential for substantial variation in the SoC of individual cells within a battery pack if cell balancing is not properly implemented. It is possible that this may result in a number of problems, including a drop in total capacity, a decline in performance, and significant dangers to safety. There are a number of factors that contribute to this phenomenon, including changes in internal resistance, uneven ageing, discrepancies in cell production, and inconsistent heating conditions. It is possible for cells to charge and discharge at very different rates, which result in imbalances as shown in Fig. 9. If the battery pack is not balanced, weaker cells might restrict the performance of the whole pack. This is because the BMS can shut down the system prematurely to preserve the weakest cell, which in turn reduces the effective capacity of the battery pack and its longevity. A dynamic adjustment of the charge distribution is made by the machine learning algorithm, which takes into account changes in cell properties, use patterns, and environmental

Table 6
Battery cell voltages (Cell 1, Cell 2, and Cell 3) using ML-based balancing and unbalanced condition.

Measuring tool	Without balancing			PA-RNN			Deep Q			Amortized Q			Adaptive NN			Automotive controller		
The initial instant of time (T_i)	Cell1	Cell2	Cell3	Cell1	Cell2	Cell3	Cell1	Cell2	Cell3	Cell1	Cell2	Cell3	Cell1	Cell2	Cell3	Cell1	Cell2	Cell3
The Final instant of time (T_f)	1s	1s	1s	1s	1s	1s	4s	4s	4s	4s	4s	4s	4s	4s	4s	4s	4s	4s
The initial instant of Voltage (V_i)	244s	244s	244s	250s	250s	250s	304s	304s	304s	244s	244s	244s	252s	252s	252s	252s	252s	252s
The Final instant of Voltage (V_f)	3.9V	3.95V	3.85V	3.72V	3.81V	3.85V	3.87V	3.87V	3.86V	3.87V	3.84V	3.83V	3.89V	3.87V	3.86V	3.89V	3.87V	3.85V
The Final instant of Voltage (V_0)	3.62V	3.86V	3.85V	3.65V	3.64V	3.64V	3.56V	3.54V	3.52V	3.59V	3.58V	3.56V	3.55V	3.56V	3.52V	3.65V	3.64V	3.62V

variables. This skill of prediction enables the system to resolve imbalances in advance, therefore assuring that all cells maintain levels of SoC that are comparable to one another. In order to improve its balancing tactics over time, the machine learning system is able to learn from past data.

Controlling and monitoring battery performance accurately relies on SoC error. Factors include inaccurate current sensing, temperature variations, cell ageing, and estimate algorithm limitations impact SoC errors. From Fig. 10, a reduced average SoC error of -1.15% was achieved by using PA-RNN, which is very effective at collecting data temporal relationships. Despite its durability and excellent prediction accuracy, it may be computationally demanding and requires a large amount of training data. On average, Deep-Q Network achieves a SoC error of -3.15% while optimizing cell balancing solutions using reinforcement learning. Although it can adjust to changing conditions, it could have more complicated implementations and slower convergence times. With a -3.25% SoC error, Amortized-Q uses amortized inference to expedite decision-making. It allows for quicker inference, although approximations mean it isn't always precise. By adjusting their structure in response to input data, Adaptive Neural Networks can achieve a SoC error of -1.04% . Though they are very accurate and flexible, they are not always easy to execute and may need a lot of resources. With a -2.65% SoC error, Automotive Controllers strike a good balance between efficiency and practicality.

Numerous issues can occur in unbalanced conditions due to its 3.5A average battery current from 0 s to 250 s as shown in Fig. 11. One issue is that it might lead to unequal cell ageing, which means that certain cells will decay at a quicker rate than others. This will reduce the battery pack's total lifetime. Second, the BMS has to restrict the total number of charge and discharge cycles to preserve weaker cells, which might lead to lower useful capacity under imbalanced settings. Finally, since certain cells are running under more stress, the danger of thermal runaway and other safety issues rises. Active cell balancing based on machine learning is used to fix these problems. Table 7 compares predictive analytic and neural network methodologies based on performance indicators for SoC balancing, energy efficiency, and other crucial requirements. The following algorithms demonstrate how this method optimizes battery currents: PA-RNN exhibits its capacity to reduce SoC faults and improve cell life by maintaining a battery current of 3.8A, capturing temporal dependencies, and providing a balanced and efficient charging and discharging process.

The fact that the Deep-Q Network can maintain a current of 3.95A demonstrates its prowess in reinforcement learning, which allows it to dynamically adjust balancing strategies for better energy distribution and smaller cell stresses. With 3.97A preserved, the Amortized-Q Network has the benefit of amortized inference, which allows for faster decision-making, which in turn leads to faster changes to balancing actions and efficient energy utilization. 4.15A current is achieved by Adaptive NN as these show their ability to adjust structures with connection weights given the input of real time data. This enables the device to offer high degrees of precision and versatility for other environments with an overall enhancement for battery life. Most importantly, the energy level at 4.32A, is regulated by Automotive Controllers, which is installed in EVs. Fig. 12 depicts an algorithm effectiveness comparison, emphasizing AQN's quick convergence, PA-RNN's correctness, ADNN's adaptability, DQN's flexibility, and AC's real-time efficiency.

6. Conclusion

This study presents an advanced solution for optimizing BMS through machine learning-based active cell balancing. A comparative analysis reveals that PA-RNN achieves the lowest average SoC error (-1.15%), outperforming Deep-Q (-3.15%) and Amortized-Q (-3.25%), while Adaptive Neural Networks further improve accuracy with an error of -1.04% . Automotive Controllers strike a balance between cost-

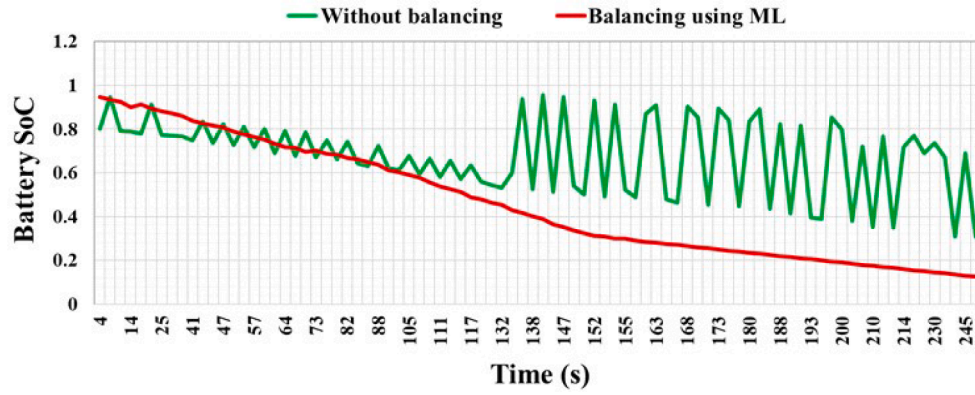
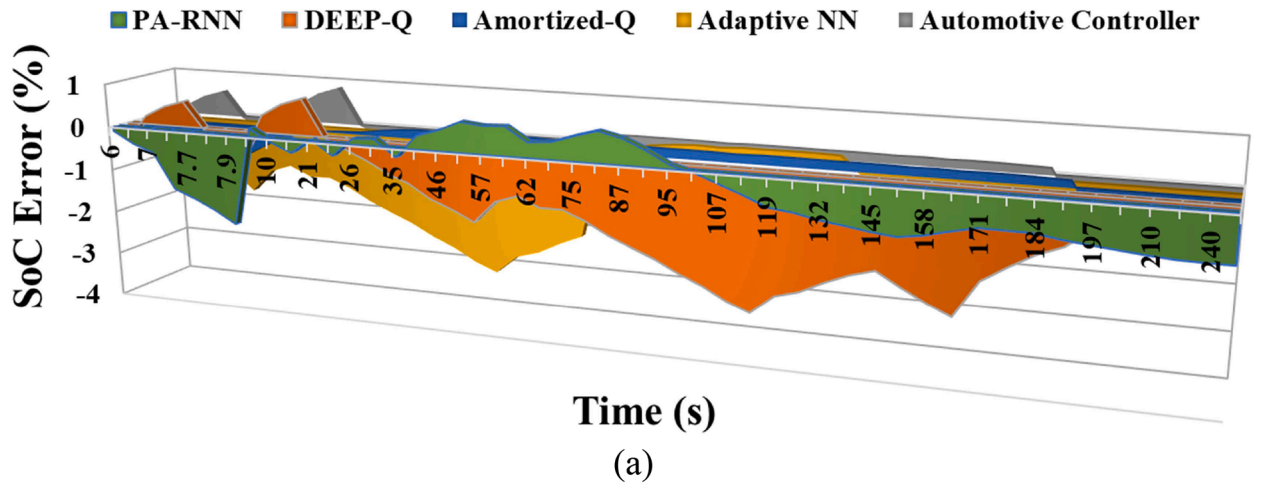
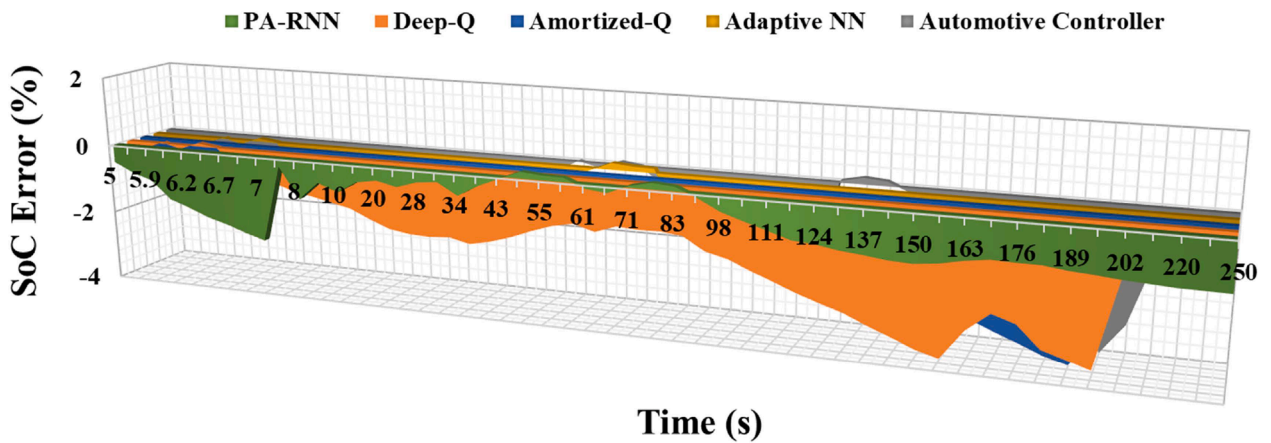


Fig. 9. Battery SoC using without balancing, and machine learning-based balancing.



(a)



(b)

Fig. 10. SoC error (%) for (a) Without balancing, (b) Balancing using PA-RNN, Deep Q-Network, Amortized Q-Learning, Adaptive Neural Network, and Automotive Controller.

effectiveness and precision, maintaining an SoC error of -2.65% . However, challenges such as inaccurate current sensing, temperature fluctuations, and uneven cell aging remain critical concerns. In terms of current optimization, PA-RNN (3.8A) minimizes SoC errors, Deep-Q (3.95A) dynamically adjusts control strategies, Amortized-Q (3.97A) enables rapid decision-making, Adaptive NN (4.15A) provides high precision, and Automotive Controllers (4.32A) are well-suited for

electric vehicle applications.

Although Amortized-Q demonstrates a slightly higher SoC error of -3.25% , Automotive Controllers strike a balance between economic feasibility and precision, maintaining an SoC error of -2.65% . Current inconsistencies, like the 3.5A charge causing uneven cell aging and capacity reduction, highlight the need for further optimization. The following machine learning algorithms offer significant improvements:

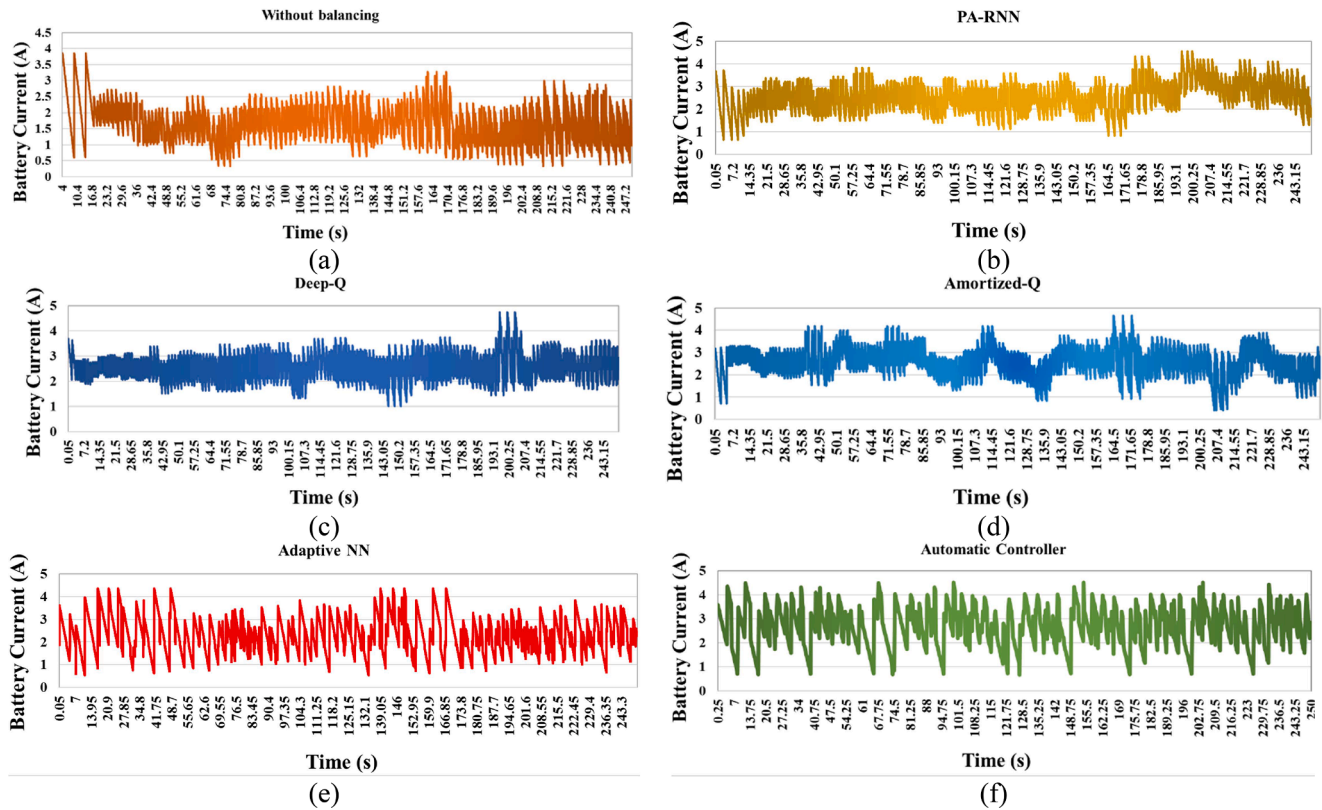


Fig. 11. Battery current in (a) Without balancing, Balancing condition using (b) PA-RNN, (c) Deep-Q network, (d) Amortized Q, (e) Adaptive Neural Network, and (f) Automatic controller.

Table 7

Comparison of predictive analytic and neural network methodologies based on performance.

Method	Description	SoC Balancing Improvement (%)	Energy Efficiency Improvement (%)	Other Key Metrics
Predictive Analytic Recurrent Neural Networks	Predicts future battery usage patterns and optimizes SoC balancing in real-time using temporal sequence analysis.	25–30 %	20–25 %	Reduced SoC convergence time by ~15 %.
Deep-Q Networks	Uses reinforcement learning to optimize energy distribution across multiple energy sources in hybrid systems.	30–35 %	25–30 %	Enhanced decision-making under uncertainties.
Amortized-Q Networks	Extends Q-networks by reducing computation time using amortization for repetitive balancing tasks.	20–25 %	15–20 %	Faster execution time by ~40 %.
Adaptive Neural Networks	Dynamically adjusts network architecture based on varying input conditions to maintain optimal energy and SoC balance.	35–40 %	30–35 %	Improved reliability under dynamic loads.
Automotive Controllers	Embedded controllers specifically designed for energy management in automotive applications, ensuring effective hybrid system operation.	20–25 %	15–20 %	Enhanced vehicle energy performance by ~18 %.

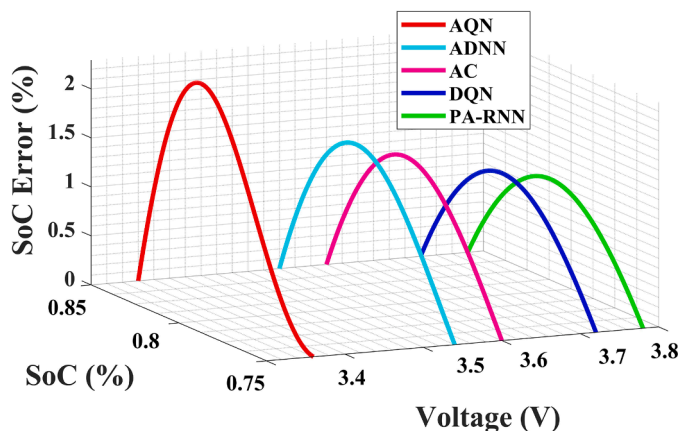


Fig. 12. Comparison of effectiveness of AQN, ADNN, AC, DQN, and PA-RNN algorithm.

PA-RNN (3.8A) minimizes SoC errors, Deep-Q (3.95A) dynamically adjusts control strategies, Amortized-Q (3.97A) supports quick decision-making, Adaptive NN (4.15A) provides high accuracy, and Automotive Controllers (4.32A) are tailored specifically for electric vehicle applications.

Future research could concentrate on integrating these ML-based balancing methods with renewable energy systems to improve sustainability, investigating compatibility with advanced battery technologies such as solid-state batteries for increased efficiency, and addressing thermal management challenges to ensure safety and performance optimization, particularly under changing environmental conditions and high-power applications [47,48].

CRedit authorship contribution statement

Veeranki Srinivasa Rao: Formal analysis. **Guna Sekhar Sajja:** Formal analysis, Software. **Vishwaraj B Manur:** Formal analysis,

Visualization. **Sairaj Arandhakar**: Conceptualization, Writing – original draft. **V.B. Murali Krishna**: Writing – review & editing, Writing – original draft, Investigation, Conceptualization.

Declaration of competing interest

The authors declare no conflict of interest

Data availability

No data was used for the research described in the article.

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